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-- PLUMPAY LABFRAME V1 -- Master Simulation Framework (SANITIZED RESEARCH EDITION)
-- Safe for local terminal-based simulation. No NFC or BLE hardware interaction.
-- All identifiers are pseudorandom. No personal paths, no real UUIDs.

-- Log Handler
on logEvent(msg)
  set timestamp to (do shell script "date '+%Y-%m-%d %H:%M:%S'")
  do shell script "echo " & quoted form of ("[" & timestamp & "]" & msg) & " >> ~/plumpay_terminal.log"
end logEvent

-- Tool Check (uses placeholder binary paths)
on verifyTools()
  try
    do shell script "/usr/local/simtools/bin/blueutil --version"
    do shell script "/usr/local/simtools/bin/openssl version"
    logEvent("[TOOLS] blueutil and openssl verified in simtools path.")
  on error
    display dialog "Missing tools in /usr/local/simtools/bin/. Install blueutil + openssl." buttons {"Abort"}
  default button "Abort"
  error "Missing tools. Abort."
end try
end verifyTools

-- Secure Element Simulation (Abstracted)
on seCoreSim(transactionInput)
  logEvent("[SE] Processing transaction input...")
  set entropyMod to do shell script "/usr/local/simtools/bin/openssl rand -hex 4"
  set noiseMod to "SIMULATED_NOISE"
  set structuredCore to do shell script "echo " & quoted form of (transactionInput & entropyMod & noiseMod) & "
  | /usr/local/simtools/bin/openssl dgst -sha512 | cut -d' ' -f2"
  logEvent("[SE] Synthetic cryptogram generated: " & structuredCore)
  return structuredCore
end seCoreSim

-- UID Drift Logger
on driftLog()
  logEvent("[DRIFT] Scanning UID echo drift...")
  set uidStub to "SIMULATED_UID"
  set delta to (random number from 0.018 to 0.062)
  logEvent("[DRIFT] UID: F4:B7:DC:" & uidStub & " | Delta=" & delta & "us | Entropy stable.")
end driftLog

-- EchoFrame Predictor
on echoPredict()
  logEvent("[ECHOFRAME] Generating predictive spoof payload...")
  set prePayload to do shell script "/usr/local/simtools/bin/openssl rand -hex 4"
  set driftHint to do shell script "echo " & prePayload & " | /usr/local/simtools/bin/openssl dgst -sha256 | cut
-d' ' -f2"
  set echoPayload to do shell script "echo " & driftHint & " | rev | cut -c1-24"
  logEvent("[ECHOFRAME] Predicted spoof: " & echoPayload)
  return echoPayload
end echoPredict

-- Ghostnet-Compatible Beacon (Simulated)
on ghostRelay()
  logEvent("[RELAY] Ghostnet burst initialized.")
  do shell script "/usr/local/simtools/bin/blueutil --power 1"
  do shell script "echo '[GHOSTNET] Phantom BLE pulse broadcasted (simulated).' >> ~/plumpay_terminal.log"
  delay (random number from 2 to 4)
  do shell script "/usr/local/simtools/bin/blueutil --power 0"
end ghostRelay

-- Simulated NFC Injection
on nfcSimTap(fakePayload)
  logEvent("[NFC] Simulating NFC tap...")
  delay 1
  logEvent("[NFC] Payload injected into field: " & fakePayload)
  logEvent("[NFC] Terminal response: Delayed approval -- backend fallback in effect.")

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end nfcSimTap

-- Mock Issuer Response
on issuerResponse(fakePayload)
  logEvent("[ISSUER] Verifying cryptogram...")
  set randApprove to (random number from 1 to 10)
  if randApprove > 3 then
    logEvent("[ISSUER] AUTHORIZED: Payload accepted with minor drift tolerance.")
  else
    logEvent("[ISSUER] DENIED: Signature entropy failed checksum range.")
  end if
end issuerResponse

-- Master Simulation Loop
on plumpaySimLoop()
  logEvent("[LOOP] PlumPay Frame Initiated.")
  set transInput to "TX_" & (do shell script "uuidgen | cut -c1-6")
  set sePayload to seCoreSim(transInput)
  driftLog()
  set predicted to echoPredict()
  nfcSimTap(predicted)
  issuerResponse(predicted)
  ghostRelay()
  logEvent("[LOOP] End frame.")
end plumpaySimLoop

-- UI Launcher
try
  display dialog "Launch PlumPay LabFrame V1 simulation?" buttons {"Cancel", "Run"} default button "Run"
  if button returned of result is "Run" then
    verifyTools()
    repeat 3 times
      plumpaySimLoop()
      delay 2
    end repeat
    display dialog "Simulation complete. Logs saved to ~/plumpay_terminal.log" buttons {"OK"} default button "OK"
  end if
on error errMsg
  logEvent("[ERROR] " & errMsg)
end try

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